

Benjamin Jauregui

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SUMMARY

PhD student in Mathematics (at Université Paris Cité – University of Chile, via a cotutelle agreement) with a strong background in mathematics and theoretical computer science. My research focuses on understanding the locality of computational problems in modern models of computation, primarily in distributed computing. More recently, I have developed interests in the intersection of distributed computing with quantum and online algorithms for discrete optimization and graph problems.

EDUCATION

PhD in Mathematics and Computer Science [IRIF, Université Paris Cité](#) **Paris, France** 2024–present

Advisor: *Pierre Fraigniaud*

PhD in Applied Mathematics [Universidad de Chile](#) **Santiago, Chile** 2023–present

Advisors: *Pedro Montealegre and Iván Rapaport*

Master in Applied Mathematics [Universidad de Chile](#) **Santiago, Chile** 2021–2022

Advisors: *Pedro Montealegre and Iván Rapaport*

Mathematical Engineering (Applied Mathematics) [Universidad de Chile](#) **Santiago, Chile** 2016–2021

PUBLICATIONS

3.1 Conferences

- Jauregui, B., Ji, J., Montealegre, P., Todinca, I. (2026). Distributed Treewidth Computation and Courcelle’s Theorem in the CONGEST Model. On Proceedings of the ACM Symposium on Principles of Distributed Computing (PODC ’26). [\[Preprint\]](#)
- Jauregui, B., Miyamoto, M. (2026). Distributed Statistical Zero-Knowledge Proofs via Sumcheck. *Brief Announcement* in Proceedings of the ACM Symposium on Principles of Distributed Computing (PODC 2’6). [\[Preprint\]](#)
- Jauregui, B., Montealegre, P., & Rapaport, I. (2025). Deterministic Distributed DFS via Cycle Separators in Planar Graphs. In Proceedings of the ACM Symposium on Principles of Distributed Computing (PODC ’25). [\[Preprint\]](#)
- Jauregui, B., Montealegre, P., & Rapaport, I. (2022). Distributed interactive proofs for the recognition of some geometric intersection graph classes. In Proceedings of the 31st International Colloquium on Structural Information and Communication Complexity (SIROCCO ’22). [\[Preprint\]](#)

3.2 Journal

- Jauregui, B., Montealegre, P., Ramirez-Romero, D., Rapaport, Iván. Compact distributed certification of geometric graph classes, Journal of Computer and System Sciences, Volume 154, 2025, 103661, ISSN 0022-0000. [\[journal version\]](#)

3.3 Preprint

- Jauregui, B., Modanese, A., Montealegre, P., Rios-Wilson, M. (2026). Strong and Hiding Distributed Certification of Bipartiteness. [\[Preprint\]](#)

TALKS

1. **Distributed Treewidth Computation and Courcelle's Theorem in CONGEST model.** PODC '26. Egham, UK. July 2026.
2. **CONGEST algorithms in sparse graph classes.** ENEDISC LaBRI Meeting. Bordeaux, France. April 2026.
3. **Distributed Courcelle's Theorem in the CONGEST model.** MoMA. Aussois, France. November 2025.
4. **Distributed Courcelle's Theorem in the CONGEST model.** Algorithms and Complexity Seminar, IRIF. Paris, France. September 2025.
5. **Deterministic Distributed DFS via Cycle Separators in Planar Graphs.** PODC 2025. Huatulco, Mexico. June 2025.
6. **Distributed DFS computation in planar graphs.** Discrete Summer School. Valparaíso, Chile. January 2025.
7. **Separator sets: Computation and application in distributed models.** LIFO Seminar. Orléans, France. October 2024.
8. **Local techniques in global problems.** Seminar on Complex Systems. Santiago, Chile. July 2024.
9. **Energy Efficient Algorithms.** PhD Seminar. Santiago, Chile. August 2023.

CONFERENCES AND WORKSHOPS

1. PODC '26. Egham, England. July 2026.
2. Modern Optimization Methods and Models in Algorithm Design (MoMA). Aussois, France. November 2025.
3. Distributed Quantum Computing Workshop. Helsinki, Finland. October 2025.
4. PODC '25. Huatulco, Mexico. June 2025.
5. DISC '24. Madrid, Spain. October 2024.

RESEARCH VISITS

Université d'Orléans. Orléans, France.

October 2024 and June 2026

IMDEA Networks. Madrid, Spain .

November 2024

U. of Bergen. Bergen, Norway.

August 2026

SKILLS AND SERVICES

Language	Spanish (native), English (advanced), French (basic)
Organizer	Organizer of the PhD seminar at Universidad de Chile.
Reviewer service	LATIN 2024, SIROCCO 2025, IWOCA 2026, PODC 2026, DISC 2026.